System design Oct 24, 2026

Performance: vs Scalability, Latency vs throughput.

Performance ~~is~~:

Scalability ⇔ Resources ↑ and performance ↑

performance problem: system slow even has 1 user.

Scalability problem. 1 user fast but slower if ↗ 10 users

Latency vs throughput.

Latency: time to finish 1 action. (1 request needs 200ms)

Throughput: the number of action in 1 time unit:
(500 requets / s)

→ target: maximal throughput and acceptable latenc

Availability; ~~consist~~ consistency
CAP theorem

CAP theorem: Read the
* consistency: √newest write OR error.

* Availability: always response for any request (but not s it is the newest)
* Partition tolerance: system still works ~~fast~~ even net is partition.

→ Trade off:

n assist with filling out this form. AutoFill Form

CP: Consistency + Partition: tolerance:

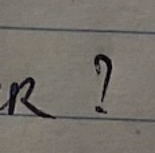
node partition: → time out. ERROR.
→ When: atomic reads and write: do all jobs at the same time. You can't break the progress

AP: Availability + Partition:

→ information is always available but not the newest
After Restoring network → it will update.

DNS _ Domain name:

DNS helps: domain → Ip address

- hierarchical: like tree 

- Authoritative: severs. the lowest level → Return ip address
- ~~Rou~~ Router / isp: (Bell, Rogers) → Resolver → caches the domain + ip.

⑥ Cache → short memories for looking Results

⑦ TTL: time to live: → ~~expired~~ expired. → deleted caches

⑧ DNS 1 propagation: delays: → Change Ip of web → wait for all caches are expired.

⑨ 4 Record

A Record: check ip for domain: domain → Ip

Ns Record: DNS ~~server~~ server answer? name server

MX Record : Mail exchange → for Recieving emails.

cName: Canonical: Nick name [www.domain.com
domain.com

Managed DNS services & Route traffic:

*) Cloudflare, Route 53 → DNE external

+) Weighted Round Robin : base on strength of servers

+) Latency-based: fastest to users.

+) Geolocation:- based: ~~based~~ canada → servers on canada

*) Disadvantages:

*) slight delay: lookup time.

*) DDos attack: